Engineering Analysis Report: 3-DOF Robotic Arm

Full Integrated Final Report (V9) - Kinematics, Dynamics & Control

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1. System Architecture & DH Parameters

Vertical height calibration: $d_1 = L_{base} + L_{offset1}$.
Effective Link 3: $L_{3e} = L_{link3} + L_{gripper}$.

$$^{0}T_{1} = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1)\cos(\alpha_1) & \sin(\theta_1)\sin(\alpha_1) & 0 \\ \sin(\theta_1) & \cos(\theta_1)\cos(\alpha_1) & -\cos(\theta_1)\sin(\alpha_1) & 0 \\ 0 & \sin(\alpha_1) & \cos(\alpha_1) & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Forward Kinematics (FK)

$$X = \cos(\theta_1)\,[L_2\cos(\theta_2) + L_{3e}\cos(\theta_2+\theta_3)]$$
$$Y = \sin(\theta_1)\,[L_2\cos(\theta_2) + L_{3e}\cos(\theta_2+\theta_3)]$$
$$Z = d_1 + L_2\sin(\theta_2) + L_{3e}\sin(\theta_2+\theta_3)$$

3. Inverse Kinematics (IK): Dual-Solution Analysis

θ_1: atan2(Y, X)

θ_3 (Elbow Up/Down):

$$D = (r^2+s^2 - L_2^2 - L_{3e}^2) / (2L_2L_{3e})$$
$$\theta_{3,up} = \mathrm{atan2}(-\sqrt{(1-D^2)}, D) \quad | \quad \theta_{3,down} = \mathrm{atan2}(\sqrt{(1-D^2)}, D)$$

θ_2: $\mathrm{atan2}(s, r) - \mathrm{atan2}(L_{3e}\sin(\theta_3), L_2 + L_{3e}\cos(\theta_3))$

4. Jacobian Matrix & Velocity Computation

To directly compute linear velocities $V = [v_x, v_y, v_z]^T$ from joint velocities $\dot{\theta}$:

$$J = \begin{bmatrix} -Y & -c_1(L_2s_2+L_{3e}s_{23}) & -L_{3e}c_1s_{23} \\ X & -s_1(L_2s_2+L_{3e}s_{23}) & -L_{3e}s_1s_{23} \\ 0 & L_2c_2+L_{3e}c_{23} & L_{3e}c_{23} \end{bmatrix}$$

Direct Substitution:

$v_x = J_{1,1}\dot{\theta}_1 + J_{1,2}\dot{\theta}_2 + J_{1,3}\dot{\theta}_3$

$v_y = J_{2,1}\dot{\theta}_1 + J_{2,2}\dot{\theta}_2 + J_{2,3}\dot{\theta}_3$

$v_z = J_{3,1}\dot{\theta}_1 + J_{3,2}\dot{\theta}_2 + J_{3,3}\dot{\theta}_3$

5. Advanced Dynamics: Torque Analysis

τ_1 (Base) includes inertial acceleration and Coriolis/Centrifugal effects:

$$\tau_1 = J_{base}(\theta) \cdot \alpha_1 + \omega_1 \cdot [d/dt\, (J_{base}(\theta))]$$

τ_2 (Shoulder) and τ_3 (Elbow):

$$\tau_2 = [\ (m_2 l_{c2} + (m_3+m_{ee}+m_p)L_2)g\cos(\theta_2)\ ] + [\ (m_3 l_{c3} + (m_{ee}+m_p)L_{3e})g\cos(\theta_2+\theta_3)\ ]$$

$$\tau_3 = [\ (m_3 l_{c3} + (m_{ee}+m_p)L_{3e})g\cos(\theta_2+\theta_3)\ ]$$

6. Singularity Analysis

Boundary Singularity occurs when $\sin(\theta_3) = 0$. Internal Singularity occurs when $\det(J) = 0$.

7. Conclusion

This comprehensive report consolidates all kinematic, dynamic, and velocity modeling required for the robotic arm's control system design.